

Antarctic Sea-Ice, Iceberg Trajectory and Navigation Decision Support

“Develop an AI/ML-enabled decision support platform capable of forecasting Antarctic sea-ice concentration, predicting iceberg trajectories, and identifying safe and fuel-efficient navigation routes for research vessels using satellite, oceanographic and meteorological datasets.”

— Ministry of Earth Sciences / NCPOR, Software category

What this document is. A complete onboarding and reference brief for the project. It assumes no prior knowledge: it explains what the system does, how it works, every technology and dataset involved, every model and what each one is for, and exactly which parts are built, which are measured, and which are still open. Read §1 and §2 to understand the project; read §3 onward as reference.

MINISTRY	Ministry of Earth Sciences (MoES) · National Centre for Polar and Ocean Research (NCPOR), Goa
DELIVERABLE	Web application with a local backend, designed to run offline aboard a vessel
TWO TRACKS	Internal hackathon prototype (8 Sept 2026) · SIH Grand Finale product (December 2026)
STATUS	Prototype trained and measured on real data · production architecture designed, 27 decisions logged, production code not yet started
HEADLINE RESULT	Sea-ice forecast beats every baseline at every horizon tested — up to +30.6 % over persistence at 7-day lead
REPOSITORY	Private · <code>SIH26059-antarctic-navigation</code>

— Contents

§1 Read this first — the project in plain English

For a brand-new team member. What problem, for whom, and what we are building.

§2 How the project works, end to end

The core idea, the pipeline, the two-tier architecture, and why it is shaped this way.

§3 Technologies involved

Every language, library, service and tool — and the ones we deliberately rejected.

§4 Data sources

Every dataset, its provider, access route, authentication, and where it is used.

§5 The models — how many, and what each one does

Five modelling components. Two are trained. Full detail on each.

§6 Results measured so far

The forecasting numbers, the routing run, and the caveats attached to both.

§7 What makes this different from the obvious build

Seven differentiators, each tied to a component and a falsifiable test.

§8 Honesty commitments and known weaknesses

What we deliberately refuse to claim, and why that is a strength.

§9 Key decisions on record

The 27 architecture decision records, summarised.

§10 Current state, open gaps and what happens next

Exactly what is done, what is not, and the phase plan to December.

§11 Repository map and how to get started

Where everything lives and what to run first.

§12 Glossary

Every acronym and domain term used in this document.

HOW TO READ THIS DOCUMENT

Sections are written so each can be read alone. Anything **measured** is stated with its number and where it was measured. Anything **estimated** says so. Anything **not yet done** is marked **OPEN**. If a claim here has no number behind it, that is deliberate — we do not have one yet, and pretending otherwise is the failure mode this project is built to avoid.

§1 Read this first — the project in plain English

Written for someone who has just joined the team and knows nothing about sea ice, ships, or this problem statement. Nothing here requires prior knowledge.

1.1 The situation

1. Antarctica is surrounded by **sea ice** — a floating layer of frozen seawater that forms, drifts, cracks and melts. It is not fixed. It can move tens of kilometres in a day during a storm.
2. India operates two permanent research stations in Antarctica: **Maitri** (Schirmacher Oasis, 70.8°S 11.7°E) and **Bharati** (Larsemann Hills, 69.4°S 76.2°E). A third, Maitri-II, is planned.
3. Once a year, during the austral summer (roughly December to April), a ship must deliver fuel, food, equipment and personnel to those stations. This is the **Indian Scientific Expedition to Antarctica (ISEA)**, run by NCPOR under the Ministry of Earth Sciences.
4. The ship currently used is the **MV Vasiliy Golovnin**, a chartered ice-strengthened Russian diesel-electric vessel. Expeditions are flagged off from Goa, but personnel fly to **Cape Town**, and the sea leg runs Cape Town → Bharati → Maitri.
5. If the ship picks a bad path, it can become **beset** — stuck in ice. That costs days or weeks, burns fuel, risks the hull, and can mean a station does not get resupplied before the window closes.

1.2 The problem we are solving

THE PROBLEM IN ONE SENTENCE

A ship's captain needs to know what the ice will look like days from now, where the icebergs will have drifted to, and which route through all of that is both safe and fuel-efficient — and they need to know it on a ship with almost no internet connection.

Each part of that sentence is a hard problem on its own:

- **Forecasting the ice.** Physics supercomputers already forecast sea ice, but their forecasts are systematically wrong in predictable places — especially at the ice edge, which is exactly where ships sail.
- **Predicting icebergs.** Icebergs drift under wind, current and the Earth's rotation. Their positions are only observed weekly, and their future positions carry large, growing uncertainty.
- **Finding a route.** The best route is not the shortest. It trades distance against ice, weather, water depth and fuel, and the constraints themselves are uncertain.
- **Doing it offline.** Geostationary satellite internet does not reliably cover south of about 70°S. Polar ships use **Iridium**, where the practical budget for a routine message is well under 50 kilobytes.

1.3 What we are building

A **decision-support platform** — not an autopilot. It advises; the master decides. It has two halves that are the same codebase running in two modes:

1. **The shore tier** runs at NCPOR on a normal workstation. It has bandwidth and a GPU. It downloads satellite and forecast data, runs the AI models, and every night builds a small compressed file called a **voyage pack**.

2. **The vessel tier** runs on a laptop on the ship's bridge. It receives the voyage pack over Iridium — or on a USB stick at port — and then works with **no internet at all**. It can recompute routes on demand in about nine seconds.

THE DESIGN RULE THAT FOLLOWS FROM THIS

Ashore: everything that needs bandwidth or a GPU. Aboard: everything that needs to answer a question. Sync: a compressed picture of the environment, never a finished answer. We deliberately do not send the ship a finished route, because then the captain could not ask "what if I go further east?" — and a system that cannot answer the operator's next question is a picture, not a tool.

1.4 The one idea at the heart of the AI

This is the single most important thing to understand about the project.

1. There is already a good physics forecast of sea ice, produced by the European **Copernicus Marine Service (CMEMS)**. It runs on supercomputers and assimilates live satellite data. We could never out-build it.
2. So **we do not compete with it. We correct it.** Our neural network does not predict sea ice. It predicts how wrong the physics forecast is going to be, and we add that correction on top.

```
corrected sea-ice concentration = clip( physics forecast + our predicted correction , 0 , 1 )
```

- Because the network outputs a correction, an untrained network outputs exactly zero and reproduces the physics forecast unchanged. **Our floor is the physics model's skill, not zero.** We can only be judged on what we add.
- The network is told, per grid cell, **how wrong the forecast is right now** — the difference between the forecast and the most recent real satellite observation. This is called the **innovation** channel, and it is what makes the whole thing work.

WHY THIS MATTERS — A REAL LESSON FROM OUR OWN HISTORY

The first version of our model saw only the physics forecast, with no satellite observations. It scored RMSE 0.1051 and **lost badly** to the dumbest possible baseline ("assume tomorrow's ice looks like today's"). Giving the model access to what the satellite actually saw is what closed the gap. We found this by measuring it ourselves, before a judge could.

1.5 What we have achieved so far

+30.6 %

BETTER THAN PERSISTENCE AT 7-DAY LEAD

1,096

REAL SATELLITE FILES USED, ZERO FAILURES

17.4

DAYS — REAL COMPUTED ROUTE, CAPE TOWN → BHARATI

~9 s

FULL ROUTING PIPELINE ON A NORMAL LAPTOP

1.6 What we are not claiming

- We are not replacing electronic navigational charts. This is decision support.
- We do not predict growlers and bergy bits — the small ice that actually holes a hull. No dataset tracks them.

- We do not print confident absolute fuel numbers, because the fuel curve we have is calibrated for a different ship.
- We do not claim immunity to climate change. We claim the system degrades visibly rather than silently.

§2 How the project works, end to end

From raw satellite files to a route on a screen. This section is the mechanical walk-through; §5 covers the models in depth.

2.1 The nine stages

#	Stage	What happens	Where it runs
1	Ingestion	One adapter per data source. Downloads satellite sea-ice observations, the physics forecast, wind, waves, sea-surface temperature, bathymetry and iceberg positions. Records a provenance row (source, version, fetch time) for every file.	Shore
2	Quality control	Before anything is combined: land-spillover masking, weather-filter false-ice detection, physical range checks, missing-swath detection, preliminary-data tagging, per-source staleness. Output is a <code>qc_flags</code> layer, not a silent cleanup.	Shore
3	Regridding	All sources are placed on one grid — NSIDC south polar stereographic, 25 km, 332×316 cells. Concentration fields use area-weighted (conservative) interpolation so ice area is preserved; forcing fields use bilinear; masks use nearest. The truth field is never resampled.	Shore
4	Ice correction	The residual U-Net ensemble reads the physics forecast plus the last 7 days of observations and emits a corrected ice field for each lead day 1–10, plus a spread.	Shore + vessel
5	Uncertainty calibration	Conformal prediction converts the ensemble spread into an empirically covered upper bound on ice concentration, per lead time and per ice regime.	Shore
6	Iceberg drift	Wagner closed-form physics projects each tracked iceberg forward, capped at 72 hours, buffered by its measured error radius.	Shore
7	Fusion	Independent ice sources are compared on QC-clean cells only. Where they disagree, the uncertainty bound widens — so the router becomes more cautious exactly where the observational record is least trustworthy.	Shore
8	Mesh + routing	meshphi builds a non-uniform adaptive mesh (finer where conditions vary). PolarRoute runs Dijkstra over it, then physics-informed smoothing. A 12-run sweep produces a dominance-filtered candidate set.	Shore + vessel
9	Pack & serve	Everything is serialised, compressed, signed and versioned into a voyage pack under 1 MB. The vessel receives it and serves a REST API to a map-based web UI.	Shore → vessel

2.2 The two-tier architecture

```

SHORE TIER ("Ops Core") — needs bandwidth + GPU —
[1] Ingestion  NSIDC CDR · AMSR2 · OSI SAF · OISST · GFS · ERA5 ·
               CMEMS forecast + GLORYS12 · GEBCO (once) · USNIC + BYU bergs
               ↓ QC pass → qc_flags layer ↓ regrid to NSIDC 25 km ↓ cache raw
[2] Training   residual U-Net × 5 · GBM drift residual (stretch)
               ↓ checkpoints — GPU ashore, never retrained aboard
[3] Nightly pack build

```

```

CMEMS forecast (base) + residual U-Net → SIC  $\mu$ ,  $\sigma$ , leads +1..+10
conformal  $\hat{q}$ (lead, regime) → SIC_upper
Wagner drift → berg positions + radii, capped at 72 h
multi-source fusion → disagreement layer
meshphi → non-uniform corridor mesh
PolarRoute × 12 sweep → dominance-filtered candidate set
[4] VOYAGE PACK (signed, versioned, < 1 MB gzip)

```

Iridium / port wifi / sneakernet USB
delta sync · resumable · integrity-checked

▼ VESSEL TIER ("Bridge")

```

same codebase, MODE=vessel, no outbound network required
[5] Local store latest pack + N previous (rollback)
[6] Local compute PolarRoute re-plan on demand (~9 s)
    U-Net re-correction when fresh observations arrive (~0.3–1 s)
[7] REST API → [8] React + Leaflet UI
    staleness banner · confidence decay · candidate-route selector
    risk-tolerance slider (labelled by MEASURED coverage)
    source-disagreement overlay · forecast-grade vs climatology-grade legs

```

2.3 Why the tier boundary sits exactly there

Three boundaries were considered and measured, not guessed:

Option	Sync payload	Verdict
(a) Vessel receives finished routes only	~1 KB	Rejected. The master cannot re-plan. Decision support that cannot answer the operator's next question is a picture, not a tool.
(b) Vessel receives the environmental mesh and routes locally	76 KB gzip (measured)	Chosen.
(c) Vessel runs ingestion and training too	GB-scale	Rejected. Impossible over Iridium.

Option (b) is only defensible because it was measured: the full PolarRoute pipeline runs in ~9 s on modest CPU, and a U-Net forward pass takes milliseconds.

2.4 The bandwidth budget — the constraint that shapes everything

Payload	Raw	Gzip	Status
Vessel-mod-elled mesh, 10°×20° box, 824 cells	551 KB	76 KB	MEASURED the representative unit
Full corridor mesh refresh	—	0.5–1 MB	ESTIMATED not measured; the mesh splits hardest in the ice edge, exactly where the corridor crosses
Daily incremental delta	—	—	UNMEASURED the load-bearing number
Environmental	336 KB	72 KB	MEASURED

Payload	Raw	Gzip	Status
mesh, pre-ves- sel-mod- el			
Route polyline + fuel/ time summa- ry	1.6 KB	796 B	MEASURED the smallest payload — quoted last on purpose
Model weights, 5 × ~3 M params fp32	—	~60 MB	never over Iridium — USB at port only

HONEST HEADLINE, STATED WORST CASE FIRST

A full corridor refresh is a minutes-to-tens-of-minutes Iridium transfer, and the daily delta — the number the design actually depends on — **has not yet been measured**. What is measured is that a representative vessel-modelled mesh block compresses to 76 KB, which makes the 50 KB daily target plausible rather than proven. A CI test fails the build if a pack exceeds budget. **If the measurement disappoints, we coarsen the mesh or lengthen the refresh cadence — we never renegotiate the budget.**

2.5 What happens when things go wrong

The system is designed around graceful degradation. It never goes silent.

Failure	Behaviour
No connectivity for days	Normal operation on the cached pack; staleness escalates green → amber → red; past pack validity it enters climatology mode — a dashed planning corridor, not a route line, with a permanent banner. Never silence.
Forecast lead exceeds validated skill	Tier down, do not refuse. Forecast-grade → CMEMS-direct → climatology-grade. The plan always covers the whole voyage; the grade of each leg is visible in the UI.
Path smoothing fails or hangs	Serve the unsmoothed Dijkstra path (0.7 s, verified) flagged un-smoothed . The fallback is inherent to the pipeline — no extra code.
Conformal coverage gate fails	The calibrated bound is withdrawn as the routed field, the risk slider is removed from the UI rather than left as theatre , and the corresponding differentiator is publicly demoted.
Ensemble members disagree strongly	The bound widens → automatically more cautious route + explicit low-confidence banner.
Sources disagree with each other	Disagreement widens uncertainty and renders as an overlay — on QC-clean cells only, so a sensor artifact cannot masquerade as genuine disagreement.

Iceberg exclusion zones cover >5 % of the corridor

Failure	Behaviour
	The berg layer auto-downgrades from hard exclusion to a weighted cost for that pack, and says so.
A data source is down at build time	Per-channel imputation policy; if the base forecast or the observations are missing, an obs-only pack ships instead — a declared pack type, not an error state.

§3 Technologies involved

Everything in the stack, what it is for, and its status. Versions marked “pinned as tested” were installed and run in this project, not read about.

3.1 Core language and runtime

Technology	Version	Role
Python	3.11.14	Everything server-side. Pinned to 3.11 because PolarRoute will not build on 3.12+ and IceNet's dependency tree requires <code><3.12</code> .
uv	—	Virtual-environment and dependency management. Chosen over pip/conda for install speed and reproducible resolution.
Node.js	—	Frontend toolchain only.

3.2 Machine learning

Technology	Role in this project
PyTorch	The residual U-Net — architecture, training loop, inference. Chosen over TensorFlow because the model is custom (a residual head with zero-initialised output) and PyTorch's imperative style makes that trivial.
NumPy	All array work: channel assembly, regridding, baseline computation, metrics.
xarray + netCDF4	Reading NetCDF climate/ocean files. Every primary dataset in this domain ships as NetCDF.
zarr	Chunked on-disk storage for the ~19–38 GB production training set. NOT BUILT
scikit-learn / gradient boosting	The optional iceberg drift residual model. Deliberately not a neural network — see §5.3.
Conformal prediction (implemented directly)	Split-conformal calibration, stratified by lead time × ice regime. Roughly 40 lines of NumPy; no library needed, and writing it ourselves means we can state exactly what it does.

3.3 Geospatial and routing

Technology	Version	Role
PolarRoute British Antarctic Survey, MIT licence	1.1.11	The routing engine. Dijkstra over a non-uniform mesh, plus physics-informed Newton smoothing of cell crossings. Written by the institution that actually operates Antarctic research vessels.
meshphi BAS, companion library	2.3.1	Builds the non-uniform adaptive mesh and hosts the data loaders. Its <code>ScalarCSVDataLoader</code> is ~15 lines — any <code>DataFrame</code> with <code>lat</code> , <code>long</code> and a named column becomes a mesh layer, so injecting our ML output needs essentially no adapter code.
GEBCO grid + TID grid	2025/26	Bathymetry (water depth) and, critically, the Type Identifier grid that distinguishes measured soundings from predicted seabed.
pyproj / EPSG:3412	—	NSIDC south polar stereographic projection, Hughes 1980 ellipsoid. The single target grid for everything.

Technology	Version	Role
geopandas / shapely	—	Iceberg exclusion polygons and route geometry.

VERIFIED, NOT ASSUMED — POLARROUTE INTEGRATION COST

Rather than reason about integration risk from a README, PolarRoute was installed and run end to end. Install: **~90 s, all binary wheels, zero compilation** — the usual geospatial-stack nightmare (GDAL, PROJ, rasterio from source) simply does not occur. Complete pipeline for a 10°×20° box: **mesh build 2.1 s → 824 cells · vessel performance 0.1 s · Dijkstra 0.7 s · smoothing 6.1 s ≈ 9 s total**. Interactive speed, demo-safe.

3.4 Backend, frontend and infrastructure

Technology	Role
FastAPI	The REST API. One service with a <code>MODE=shore vessel</code> switch — two deployment modes of the same application, not two products. This roughly halves the build and is architecturally honest, because the routing and inference code genuinely is identical on both sides.
React + Leaflet	The map-centred operator UI. Every response is annotated with its age and confidence.
GitHub Actions	Two jobs: the daily CMEMS forecast harvest (06:00 UTC — running now, and the data it collects cannot be back-filled), and the sync-budget CI test that fails the build if a pack exceeds its size budget.
Kaggle Notebooks	Free T4 GPU for training — 30 GPU-hours/week, and the whole 5-member ensemble is costed at 8–12 GPU-hours. The prototype was trained here.
Local RTX 5060	Development training and iteration.
<code>tar.zst</code> + signing	Voyage pack format: compressed, signed, versioned, delta-syncable, rollback-capable.
<code>copernicusmarine</code> , <code>cdsapi</code>	Official Python clients for CMEMS and the Copernicus Climate Data Store.

3.5 Technologies deliberately rejected

Recorded because the reasoning is part of the deliverable.

Rejected	Why
LLM / RAG / agent framework	There is no natural-language task in this problem statement. Every component is a numerical prediction or a graph search. Deterministic and classical methods are better suited here, cheaper, verifiable against held-out data, and able to run offline on a ship. A chatbot over real data is precisely the convergent shallow build we are differentiating from.
The icenet library	VERIFIED Its pins (<code>tensorflow<2.16</code> , <code>netcdf4<1.6.1</code> , plus the deprecated <code>motuclient</code> , 51 dependencies) directly conflict with <code>meshphi/PolarRoute</code> , which resolve to <code>netcdf4==1.7.4</code> . The two cannot share a Python environment. Also: IceNet's published validation is Arctic, and no Southern-Hemisphere weights exist — it would need full retraining anyway.
Reimplementing grid A*	The measured integration cost of PolarRoute is low, so the usual justification evaporates. Reimplementing would forfeit non-uniform mesh-

Rejected	Why
	ing, physics-informed smoothing and the vessel-performance framework — and invite the question "why did you ignore the field's own open-source tool?"
CICE / LIM2 dynamical ice model	Operational infrastructure run by national centres. Not standable-up by a hackathon team — and our design now consumes the output of exactly such a system instead of pretending to rebuild one.
Quantum-annealing route optimisation	Real in the current literature, but requires D-Wave access and buys optimisation speed we do not need at ~9 s per route. Complexity with no benefit at our scale.
LSTM on the ice-extent time series	Predicts a scalar. The router needs a spatial field.
A neural network for iceberg drift residual	The residual problem is small-data and tabular. Gradient boosting is the correct tool for that regime; feature importances also let us demonstrate the model recovering known physics, which is worth more to a judge than a marginally better score.

§4 Data sources

Every dataset was checked live from a sandbox shell on 31 August 2026, both by HTTP fetch and by `curl`. No synthetic or placeholder data is used anywhere in the product — that is a hard project rule.

PROJECT RULE: NO DUMMY DATA, EVER

Every dataset used for training or demonstration, and every model inference, is real. Where real data was not yet available, work stopped and a source was found rather than scaffolding with synthesised placeholders. The only permitted exception is minimal synthetic fixtures inside unit tests that check code plumbing, not model behaviour.

4.1 Master table — every source

Source	Provider	What we take from it	Auth	Status
NOAA/NSIDC Sea Ice CDR G02202 v4/v6, G10016 NRT	NSIDC / NOAA	The truth field. Observed sea-ice concentration, 25 km south polar stereographic, 332×316. Passive microwave SSM/I-SSMIS, 1979–present including a near-real-time stream. Also carries <code>qa_flag</code> and <code>stdev</code> , which feed QC for free.	None	1,096 FILES DOWN-LOADED, 0 FAILURES
CMEMS GLORYS12 GLOB-AL_MULTIYEAR_PHY_001_030	Mercator Ocean / EU Copernicus	The back-ground field for training. Ocean reanalysis, 1/12° (~8 km), 50 depth levels, 1993–present, daily. Provides <code>siconc</code> (ice concentration) and <code>sithick</code> (thickness).	Free account	USED IN PROTOTYPE
CMEMS operational forecast GLOB-AL_ANALYSISFORECAST_PHY_001_024	Mercator Ocean / EU Copernicus	The back-ground field in production — the real 10-day	Free account	DAILY HARVEST CRON RUNNING

Source	Provider	What we take from it	Auth	Status
		operational forecast we bias-correct. Also currents (u/v), SST, salinity, sea-surface height, mixed-layer depth.		
AMSR2 NSIDC-0803 / AU_SI12	JAXA / NSIDC	Independent second ice observation at 12.5 km — used for the disagreement layer and high-resolution ice-edge cross-check, deliberately not as a model input so its independence is preserved.	Earthdata login	PLANNED
EUMETSAT OSI SAF OSI-401-d, 408-a, 402-d, 403-d, 405-d	EUMETSAT	Third ice voice: concentration (10 km), ice edge, ice type, and low-resolution ice drift (62.5 km). Ice type matters for POLARIS (§10.4).	Free account	PLANNED
ESA CCI Sea Ice	ESA / CEDA	Long-baseline concentration CDR at 25/12.5/50 km, 1991–2020+. Training and validation depth.	Some open, some CDS	PLANNED
ERA5 reanalysis-era5-single-levels	ECMWF / Copernicus C3S	Historical forcing for training: 10 m u/v wind, 2 m temperature, mean sea-level pressure, and — from the same dataset — significant wave height, mean wave direction and	Free CDS account	ENDPOINT VERIFIED

Source	Provider	What we take from it	Auth	Status
		period. 0.25°, hourly, 1940–present.		
NOAA GFS (NOMADS)	NOAA NCEP	Operational wind forecast for serving — global 0.25°, 4 cycles/day, out to 384 h. Region-sub-settable so an Antarctic wind-only request is small.	None	LIVE QUERY EXECUTED
NOAA OISST v2.1	NOAA NCEI	Daily sea-surface temperature, 0.25° global, since 1981. Ships a <u>preliminary</u> flag we propagate rather than ignore.	None	CURRENT TO 2 DAYS AGO
USNIC Antarctic Icebergs	US National Ice Center	Current ice-berg positions and sizes, weekly, as CSV and shapefile — immediately GIS-usable. Tracks bergs ≥20 sq nm or ≥10 nm longest axis.	None	VERIFIED LIVE
BYU/NIC Iceberg Database	Brigham Young Univ. + USNIC	Historical ice-berg training data: daily lat/lon, major/minor axis, 1978–2025. Plus a statistical database with rotation velocity and surrounding-environment mask.	None	VERIFIED LIVE
GEBCO gridded bathymetry + TID grid	IHO + IOC (Seabed 2030)	Water depth at 15 arc-second (~450 m) resolution, and the provenance grid saying which cells	None	NOT YET DOWNLOADED

Source	Provider	What we take from it	Auth	Status
		are surveyed versus pre-dicted.		
Sentinel-1 SAR (CDSE)	ESA / Copernicus	C-band radar imagery — works through Antarctic winter darkness and cloud. Ice edge, lead detection, iceberg detection.	Free CDSE to-ken	STRETCH GOAL ONLY
NCPOR NPDC + Synoptic-Bharati	NCPOR (India)	Automatic weather station and station observation data from Maitri and Bharati — local ground truth near India's own stations , and a strong point for an MoES audience.	Varies	STRETCH — VALIDATION ONLY

4.2 The one genuine data gap

THERE IS NO “ANTARCTIC SHIPPING LANE” DATASET — AND THERE NEVER WAS

No downloadable recommended-ice-lane dataset exists from the IHO or anyone else. Official Electronic Navigational Charts for Antarctic waters are licensed and paywalled through national hydrographic offices, and Antarctic coverage is notoriously sparse because the region is barely surveyed. The IMO Polar Code and METAREA products are safety-broadcast systems, not structured route datasets.

This is not a failure in our research — it is a confirmation that route computation is the product's actual job, not an input it should be looking to ingest.

4.3 A related honesty note on bathymetry

Much of the Antarctic corridor's GEBCO bathymetry is predicted from satellite altimetry, not surveyed by echo sounder. A router that treats depth as truth is trusting a model with metre-to-hundreds-of-metre error. We handle this rather than merely admitting it:

- We ingest GEBCO's **Type Identifier (TID) grid** and carry it as a mesh layer.
- The depth margin is **conditional on provenance**: minimum under-keel clearance of **20 m in surveyed cells** and **max(50 m, 3 × draft) in predicted cells**.
- A “predicted bathymetry” overlay shows the master which parts of a route cross unsurveyed seabed, and the disclosure sits next to the depth legend in the product — not only in a document.

§5 The models — how many, and what each one does

DIRECT ANSWER TO THE QUESTION

The system has **five modelling components**. **Two of them are trained** — one deep-learning model (load-bearing) and one classical-ML model (optional). The other three are physics or exact algorithms with no learned parameters. There are also two supporting layers that are not models but shape every output: conformal calibration and multi-source fusion.

#	Component	Trained?	Type	Job in one line
1	SIC bias-correction residual U-Net x 5	Yes — deep learning	Convolutional neural network	Make the physics sea-ice forecast more accurate.
2	Wagner iceberg drift	No — closed-form physics	Analytical equation	Predict where each iceberg will drift.
3	Iceberg drift residual (GBM) STRETCH	Yes — classical ML	Gradient-boosted trees	Correct the physics drift model's systematic error.
4	Vessel performance model	No — published physics	Semi-empirical formulas	Convert ice, wind and waves into achievable speed and fuel burn.
5	Route optimiser	No — exact algorithm	Dijkstra + Newton smoothing	Find the best path through the resulting cost field.
—	Conformal calibration layer	Fitted, not trained	Statistical	Turn model

#	Component	Trained?	Type	Job in one line
				spread into an honest, measured uncertainty bound.
—	Multi-source fusion layer	No	Architectural	Compare independent ice sources; widen caution where they disagree.

5.1 Model 1 — Sea-ice concentration bias-correction residual U-Net

TRAINED & MEASURED

This is the load-bearing AI of the whole system. Everything else consumes its output.

What it does

It reads a physics sea-ice forecast plus the last week of real satellite observations, and outputs a **correction field** — a per-grid-cell number that gets added to the forecast.

```

ΔSIC_k = g_θ( F_k , O_{t-6..t} , F_0 - O_t , forcing , k )
SIC_k  = clip( F_k + ΔSIC_k , 0 , 1 )

F_k      the physics forecast valid k days ahead
O_{t-6..t} the last 7 days of real satellite observations
F_0 - O_t the INNOVATION — how wrong the forecast is right now, at this cell
k        the lead time, fed in as its own input channel

```

Why a residual, and not a direct forecast

1. **The floor is CMEMS's skill, not zero.** $\Delta\text{SIC} = 0$ reproduces the operational forecast exactly. The model cannot be much worse than the bar, and can only be judged on what it adds.
2. **The innovation channel targets a documented, specific defect.** Copernicus is known to systematically underestimate ice concentration in the marginal ice zone. Anchoring the correction to the current per-cell error is the mechanism by which the model can learn that specific bias rather than a generic climatological offset.
3. **The model can be small.** A residual on an already-skilful field needs far less capacity than a forecaster — ~3 M parameters instead of 8–15 M, which is what makes the whole ensemble fit inside free-tier GPU hours.

THIS WAS A CORRECTED MISTAKE, AND THE CORRECTION IS INSTRUCTIVE

Revision 1 of the design committed to a from-scratch direct forecaster competing with CMEMS. An adversarial review killed it: our own competitive analysis had already called CMEMS “the operational baseline any novel forecasting claim has to beat, and it's a hard bar,” and then we set our bar at naive persistence instead. **We had ducked our own bar.** The gap analysis had even named the right method — “DL bias-correction on top of a short-range ice-ocean model” — but the example given was TOPAZ4, an Arctic system, so it was filed as unusable. That was reading a method as a product. The method transfers to Antarctica; the global CMEMS system was already sitting in our own data inventory, being used for currents and ignored for ice.

Architecture

Property	Value
Shape	3-level U-Net — encoder / bottleneck / decoder with skip connections. Each block: Conv3×3 → BatchNorm → ReLU, twice.
Input (production)	19 channels — see the table below
Input (prototype)	12 channels — the subset obtainable from data already on disk
Output	1 channel — Δ SIC, deliberately unbounded. Clipping to a valid concentration happens after adding the base forecast, because it is the sum that must land in [0,1], not the correction alone.
Parameters	Measured at depth 3, 19 channels: <code>base_filters=16</code> → 485,041 · 24 → 1,088,137 · 32 → 1,931,617 · 40 → 3,015,481 (default) . The trained prototype has 1,929,601 .
Output head	Zero-initialised (<code>nn.init.zeros_</code> on both weight and bias). An untrained model is therefore a strict no-op that reproduces the forecast exactly, rather than injecting random noise into an already-skilful field.
Grid handling	The 332×316 polar-stereographic grid is not divisible by 2^3 , so the input is replicate-padded up to a multiple and cropped back at the end, keeping skip connections aligned.
Ensemble	5 members , varied by random seed and input perturbation — forcing channels jittered at the analysis-error scale, and a resampled observation window. Seed-only ensembles capture parameter uncertainty alone and systematically under-disperse.
Lead handling	Lead time is an input channel, not an output head. One network serves all leads 1-10, and the training set is 10× larger per forecast cycle.

The 19 production input channels

Ch	Channel	Source
1	Base forecast SIC at lead k	CMEMS operational forecast (serve) / GLORYS12 (train)
1	Base SIC at lead 0 (the analysis)	as above
7	Observed SIC, previous 7 days	NOAA/NSIDC CDR, 25 km
1	Innovation $F_0 - 0_t$	derived — the key channel
3	2 m temperature, 10 m u/v wind	ERA5 (train) / GFS (serve)
1	Sea-surface temperature	NOAA OISST v2.1

Ch	Channel	Source
2	Day-of-year sine and cosine	derived — seasonality
1	Lead time k, normalised	derived
1	Land / ocean mask	derived from GEBCO
1	QC-flag fraction	our own QC pass — so the network can learn to distrust what the ingest layer distrusted

The prototype's 12 channels are channels 1, the 7 observations, the innovation, the two day-of-year terms and the mask — everything obtainable without ERA5 or OISST.

How it is trained

Training is deliberately split in two stages, because bias-correcting a forecast requires an archive of past forecasts — and CMEMS keeps only a rolling window, so a multi-decade archive of its lead-k fields does not exist to be downloaded.

Stage	Base field	Period	Purpose
A bulk training	GLORYS12 reanalysis SIC as a model background, paired with NSIDC CDR observations as truth	1993–2018 train 2019–2021 validate 2022–2024 test 2023 reported separately	Learn the model-minus-observation error structure of the Mercator/ NEMO-LIM ice model that the CMEMS forecast is built on — same model family, same biases, deep archive.
B transfer + acceptance	The real CMEMS 10-day forecast, harvested one cycle per day	Sept 2026 onward, accumulating	Fine-tune on true forecast statistics, and verify Stage

Stage	Base field	Period	Purpose
			A trans-fers.

THE TIME-CRITICAL DEPENDENCY EVERYONE ON THE TEAM SHOULD KNOW

The Stage B archive cannot be back-filled. Every day the daily CMEMS harvest does not run is a day permanently missing from the acceptance test — which is the strongest claim in the entire pitch ("we beat the operational forecast on its own held-out output"). The cron is running now. It must not be switched off.

Compute budget

Quantity	Value
Forward + backward cost	~8 GFLOP/sample
Samples per epoch	9,500 cycles × 10 leads, subsampled to 3 leads/cycle ≈ 28,500
Cost per epoch	~230 TFLOP
On one NVIDIA T4	~2-4 min/epoch → 40 epochs ≈ 1.5-2.5 h/member → 8-12 GPU-hours for all five
On CPU only	~85 h for the ensemble — rejected , leaves no room to iterate, and iteration is where calibration gets fixed
Inference	5 members × 10 leads over the corridor ≈ 15 GFLOP → 0.3-1 s on a bridge laptop, no GPU. This is what makes the two-tier claim hold.
Training data volume	19 channels × 9,500 days ≈ 38 GB at float16, or ~19 GB with SIC as scaled uint8. Chunked zarr on local SSD.

How it is evaluated — and the gate that can kill it

- **Strictly temporal split.** Never random. Sea ice barely changes overnight, so a random split puts adjacent days on both sides and produces a spectacular, meaningless score. This is the most common fatal flaw in ML-on-climate submissions and is stated explicitly for that reason.
- **Baseline 1 — raw packed CMEMS. Mandatory and build-blocking.** For each lead, the corrected forecast must beat the raw forecast on both IIEE (integrated ice-edge error) and CRPS, on held-out real forecast cycles. **A lead that fails is not claimed** — the pack ships the raw field for that lead labelled `uncorrected`, and the declared skill horizon shortens.
- **Baseline 2 — damped anomaly persistence** as the floor.
- **Baseline 3 — the trivial uncertainty proxy.** The calibrated bound must beat free inter-source disagreement as an uncertainty estimator. If a three-way difference predicts the error as well as the trained σ does, the trained σ is decorative and we say so.
- **Metrics:** RMSE, IIEE at the 15 % threshold (the navigationally meaningful one), binary F1 at the vessel-relevant threshold, CRPS, interval score.
- **Stress test: 2023**, the Antarctic extreme-minimum year, evaluated and reported separately, with no threshold — publication is the requirement, not passing.

5.2 Model 2 — Iceberg drift, Wagner (2017) closed form DESIGNED

Load-bearing, and deliberately **not** machine learning.

$$\mathbf{v}_i = \mathbf{v}_w + \gamma(\alpha \cdot \hat{\mathbf{k}} \times \mathbf{v}_a + \beta \cdot \mathbf{v}_a)$$

$\mathbf{v}_i$ iceberg velocity $\mathbf{v}_w$ ocean current velocity
 $\mathbf{v}_a$ wind velocity $\hat{\mathbf{k}}$ vertical unit vector (Coriolis term)
 α, β, γ functions of berg dimensions and latitude

- **Inputs:** ocean current (CMEMS), wind (ERA5 historical / GFS operational), berg length and width (BYU/USNIC), latitude (Coriolis).
- **No fitted parameters** beyond standard literature drag coefficients.
- **Why it is the baseline:** zero training data required, closed-form (so no ODE integration and therefore no numerical-stability failure mid-demo), implements in ~100 lines, unit-testable against the published reference, and it is the physics core that IDRIFTNET — the 2025 state of the art — itself builds on. It is not a strawman baseline; it is the field's base case.
- **Evaluation:** leave-one-iceberg-out. Segments from the same berg are heavily correlated, so a random segment split leaks the same trajectory into both sides and is meaningless. Metrics are Average and Final Displacement Error at 24 / 48 / 72 h.

THE HORIZON CAP, AND WHY IT EXISTS

Physics-only displacement error is **127-147 km** at 7 days. An honest uncertainty radius that size around every tracked berg turns the hazard layer into either an ocean-wide blockade (honest but useless) or a quietly shrunk buffer (useful but dishonest). Neither is acceptable. So: **drift is projected 72 hours, never further**. Beyond that we draw nothing and show last-known positions labelled “position as of <date>, not projected” — a stale fact, not a fabricated forecast. And if exclusion polygons ever cover more than **5 % of navigable corridor area**, the layer auto-downgrades from hard exclusion to a weighted cost, and the UI says so.

5.3 Model 3 — Iceberg drift residual, gradient boosting STRETCH GOAL

- **What it does:** predicts the Wagner model's 24-hour displacement residual ($\Delta x, \Delta y$) from tabular features — berg major/minor axis, local wind speed and direction, current speed and direction, local ice concentration, latitude, distance from coast.
- **Why gradient boosting and not a neural network:** the residual problem is small-data and tabular — BYU/USNIC give weekly-to-daily positions for large tabular bergs only. Gradient boosting is the correct tool for that regime. IDRIFTNET's own stated motivation for going hybrid was that pure neural approaches converge poorly on the available data; the same logic pushes further toward classical ML for the correction term.
- **Interpretability is a judging asset here.** Feature importances let us demonstrate that the model recovers known physics — wind dominating for smaller bergs, currents dominating for large tabular ones. A model that rediscovers the physics is far more convincing than one that merely scores well.
- **It must beat both baselines to ship** — velocity persistence and pure Wagner. If it does not, the physics baseline ships alone, and that is a perfectly good outcome.

The fallback is specified as a rule, not an intention, and is unit-tested:

```
fire_fallback(x, Δ) = ANY_OOD(x) OR |Δ| > c_sanity
```

```

ANY_OOD(x) : for any feature j,  x_j < min_j - 0.10·range_j
              or x_j > max_j + 0.10·range_j
              (ranges taken from the TRAINING fold only, frozen at fit time
               and serialised into the model artefact next to the weights)
c_sanity   : p99 of |residual correction| over the training fold, in km

```

Per-feature ranges beat Mahalanobis distance or an isolation forest here because they are inspectable in the UI (“berg B-15 fell outside the trained wind-speed range”) and cannot themselves be miscalibrated. Two unit tests assert the fallback actually fires — because an untested fallback is an unfired fallback.

5.4 Model 4 — Vessel performance model REUSED, NEEDS RECALIBRATION

Not trained. Published semi-empirical naval-architecture formulas, taken from PolarRoute's own vessel class and re-parameterised for our ship.

Sub-model	What it computes	Status
Ice resistance	Froude-number formula using ice concentration, thickness, density, hull beam and hull type. Slender-hull parameters [4.4, -0.8267, 2.0] ; blunt [16.1, -1.7937, 3] .	Works
Speed inversion	If ice resistance exceeds the hull's force limit, solve for the safe speed that brings it back under.	Works
Wind resistance	Real 8-heading apparent-wind model on ERA5 u_{10} / v_{10} . Feeds resistance and therefore fuel.	Works
Fuel	Polynomial in speed and resistance, returning tonnes/day.	CALIBRATED FOR THE WRONG SHIP
Wave resistance	Kreitner formula — defined in three PolarRoute vessel classes and called by none of them.	DEAD CODE — SEE §10.4

WHY WE WILL NOT PRINT AN ABSOLUTE FUEL NUMBER

PolarRoute's fuel polynomial is empirically fitted to the **RRS Sir David Attenborough**, a British ship. We do not have the Vasilii Golovnin's power/fuel curve. Therefore fuel is reported as a **relative comparison between candidate routes** — which is what actually informs the decision — and any absolute tonnes/day figure is labelled uncalibrated model output. Competitors will display confident absolute fuel savings they cannot possibly have calibrated. Declining to do so is a differentiator, not a shortfall.

Similarly, ice **thickness and density** come from meshphi's lookup table, whose own docstring reads “Creates a simulated dataset ... based on scientific literature.” It is a literature-derived climatological parameterisation of the kind used throughout the field — not fabricated data, but it must be labelled, because the ice-resistance figure inherits its uncertainty. Note also that **without thickness and density present in the mesh, the ice-resistance speed adjustment silently does nothing** — verified in source, and a live trap for anyone wiring this up.

5.5 Model 5 — Route optimiser RUN ON REAL DATA

- **Algorithm:** Dijkstra over the non-uniform adaptive mesh — exact on that mesh, not a heuristic — followed by **Newton-method crossing smoothing**, which relaxes cell-boundary crossing points the way light refracts, turning a kinked shortest path into a physically sailable one.

- **Objectives available:** `traveltime`, `fuel`, `battery`.
- **Multi-objective handling.** PolarRoute accepts only a single objective string, so we sweep: **2 objectives × 6 risk-tolerance settings = 12 runs, ~7 s each ≈ 84 s**. Every resulting route is then scored on all three quantities — passage time, modelled fuel, and route-integrated ice risk — so the trade-off triple is measured on the route, not inherited from the objective it was solved under. Dominated routes are discarded.
- **Named honestly:** we call it a dominance-filtered candidate set (12-run sweep), not a “Pareto frontier” — 12 samples of a continuous trade-off surface is a set, not a frontier. If the filter leaves one route, we show one route and say the trade-off collapsed.

5.6 The two supporting layers

Conformal calibration

```
score s_i = ( y_i - μ_i ) / ( σ_i + ε )           one-sided: we bound only from above
q̂(stratum) = [(n+1)(1-α)] / n  empirical quantile of { s_i } within the stratum
SIC_upper = clip( μ + q̂(lead, regime) · σ , 0 , 1 )
```

- **Distribution-free** — makes no Gaussian assumption about a variable bounded in [0,1] whose error at the ice edge is strongly asymmetric.
- **Stratified into 30 strata:** 10 lead times × 3 regimes (open water <15 %, marginal ice zone 15-80 %, pack >80 %). Miscalibration is worst in the MIZ at long leads — exactly where the router consumes the bound.
- **The operator control is target coverage** (80/90/95 %), not a sigma multiple. The demo line changes from “this is 1.28 sigma” to **“this says 90 % and it measured 90.4 % on held-out data.”**
- The ensemble supplies the spatial shape of the uncertainty; the conformal quantile supplies its magnitude, from measured residuals.

PRE-COMMITTED PASS/FAIL — WRITTEN BEFORE THE NUMBERS EXIST

At nominal 90 % coverage, on held-out 2022-24 data: empirical coverage in MIZ cells at lead 5 must fall in **[85 %, 96 %]** — over-coverage fails too, because a bound that is right 99 % of the time is too wide to steer by — and mean bound width at lead 5 must be **under 25 SIC-%**. **If any check fails**, the bound stops being the routed field, the slider is removed from the UI rather than left as theatre, and the corresponding claim is publicly demoted. This branch is written down now so it cannot be quietly renegotiated afterwards.

Multi-source fusion and disagreement

- Where two or more independent, QC-clean sources cover a cell, inter-source disagreement is retained as **its own mesh layer**.
- Disagreement **widens the uncertainty bound**, so the router is automatically most cautious exactly where the observational record is least trustworthy. It changes the route — it is not a second basemap.
- QC runs first. Without it, a fusion layer cannot tell a sensor artifact from genuine disagreement, and would widen the route around a phantom. Flagged cells are excluded from the disagreement statistic and rendered distinctly — they are different facts.

§6 Results measured so far

Everything below is from the ISIH prototype, trained 31 August 2026 on a Kaggle T4 GPU. These are the numbers to quote.

6.1 Sea-ice forecasting

Lead time	Our model	Persistence	Background + pixel bias	Raw GLORYS12	Gain vs persistence
1 day	0.0465	0.0570	0.1461	0.1630	+18.5 %
3 days	0.0720	0.0958	0.1461	0.1630	+24.9 %
5 days	0.0938	0.1204	0.1468	0.1637	+22.1 %
7 days	0.0968	0.1395	0.1468	0.1637	+30.6 %

RMSE in sea-ice-concentration fraction; lower is better. 1,929,601 parameters, 12 input channels, one model per horizon, same seed at every lead so the horizons are comparable. Test set scored exactly once.

THE OPERATIONAL CLAIM THIS SUPPORTS

Our error grows more slowly than the baseline's. From day 1 to day 7 our RMSE rises **2.1×** while persistence rises **2.5×**. The advantage is largest at 7 days — which is the horizon a voyage is actually planned over.

“The further ahead the ship needs to plan, the more the model matters.” That is an operational claim, not a benchmark score.

Why these numbers are trustworthy

- **Three-way temporal split.** Train 70 % → validation 15 % (used only to pick the epoch) → test 15 %, **scored exactly once**. An earlier run reported +27 % by selecting the best of 30 epochs on the same set it then reported — that number was inflated and is not used.
- **Split by time, never randomly.** A random split would put adjacent days in both train and test and produce a spectacular, meaningless score.
- **Persistence is not a strawman.** Measured on real 2020 NSIDC data it scores 0.0359 at same-day lead — better than any background correction. Had we evaluated at same-day lead, persistence would have won and the honest conclusion would have been that the model adds nothing.
- **Baselines were computed before training**, so the bar was fixed before the result was known.
- **Real data throughout.** 1,096 NSIDC files, zero failures.

6.2 Routing

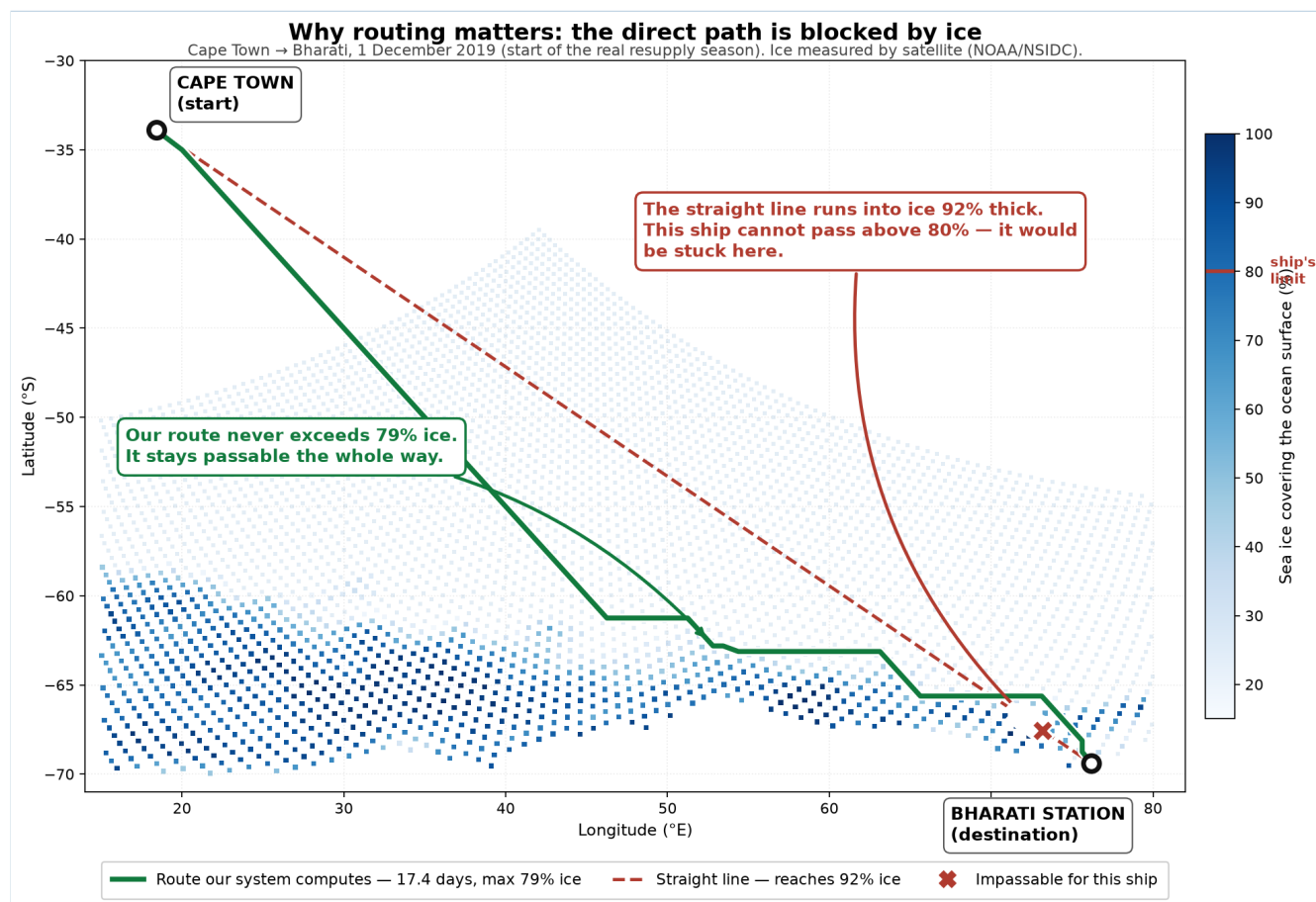


Figure 1. Real computed route, Cape Town → Bharati, on real satellite ice for 1 December 2019, using PolarRoute 1.1.11. The figure computes its own annotation numbers at render time, so the claims cannot drift from the data if the route or date changes.

Quantity	Value
Voyage time	17.4 days
Waypoints	59
Maximum ice on our route	79 %
Maximum ice on the straight line	92 %
Vessel ice limit as configured	80 %
Fraction of the straight line that is impassable	3.7 %

A CLAIM WE HAD TO WITHDRAW — AND THIS IS THE CULTURE WE WANT

The first version of this figure said the route “bends around the thickest ice.” Measuring showed that is **false**: the time-optimal route crosses more average ice than the straight line (19.6 % vs 8.4 %), because it trades moderate ice for a shorter path. Only the maximum matters, and there the route genuinely wins.

So if asked “does your route avoid ice?” the honest answer is: **it avoids impassable ice, and accepts moderate ice where that is faster**. That is what optimising travel time under a hard ice constraint actually means, and it is a better answer than a vague claim about avoidance.

6.3 Caveats attached to these results — state them, do not hide them

1. **The background is a reanalysis, not a forecast — the sharpest question we will be asked.** Raw GLORYS12 scores 0.1630 at lead 1 and 0.1637 at lead 7 — essentially flat. A real forecast degrades with lead time; this does not, because a reanalysis assimilates satellite observations including near the date being predicted. The honest answer: if GLORYS12 contained the answer its own RMSE would be near zero rather than 0.163, which is worse than persistence at every horizon — so it is not handing over the target. But it may carry some future signal, and we do not claim otherwise. The clean test is the ablation below, and production corrects a real forecast where this cannot arise.
2. **Validation scores better than test at every lead** (lead 3: val 0.0611 vs test 0.0720). Expected — validation selects the epoch, so it is optimistically biased. **Always quote the test numbers.**
3. **Single model per horizon, no uncertainty layer yet.** Production trains five and turns their disagreement into a calibrated safety margin that changes the route.
4. **12 channels, not 19.** No ERA5 wind/temperature, no OISST — the physical drivers of ice motion are currently invisible to the model.
5. **~2 years of training data** (2019–2020) versus 1993–2024 for production.
6. **Corridor region only** (5°W–85°E, 48°S–80°S), not circumpolar.
7. **Routing has no bathymetry.** GEBCO was not downloaded; the mesh reported “no elevation data” for every cell. Shallow water and land are not being avoided. This must be added before any claim about navigational safety.
8. **Routing used observed ice, not forecast ice.** Wiring the trained model into the router is what turns this from a hindsight map into a decision-support tool.

THE ABLATION THAT SETTLES CAVEAT 1 **NOT YET RUN**

Retrain with the background channel zeroed, leaving only real past observations and derived channels. **If the score barely moves** — the background contributes little, leakage is a non-issue, and the model's skill comes from observation history. Strongest possible answer. **If the score drops sharply** — the background is doing real work, and we must state that the prototype's numbers may not transfer to a live-forecast setting until production tests it. Either outcome is worth having before a viva. Cost: about 20 minutes.

§7 What makes this different from the obvious build

Given only a one-sentence problem statement, every competent team converges on roughly the same architecture. Knowing what that is — and where it is shallow — is how we differ.

7.1 What a strong competing team plausibly builds

1. Pull CMEMS ice concentration + NIC/BYU iceberg positions + ERA5 wind/current data.
2. Fine-tune or reimplement a U-Net for short-horizon ice forecasting.
3. Iceberg trajectory as either pure physics or an LSTM — rarely both.
4. Routing as A*/Dijkstra with ice as an edge-cost penalty — a static shortest path.
5. Wrap it in a Leaflet dashboard with a chat layer, demo three cherry-picked scenarios.

We know this is the convergent pattern because it already happened: for a sibling MoES problem statement in SIH 2025 (“FloatChat”), **at least four teams independently built the same chatbot-over-real-data wrapper**, where the “AI” was retrieval and NL-to-SQL translation rather than a trained predictive model.

7.2 The seven differentiators

D1 — Uncertainty reaches the routing decision, not just the legend

We route on a probabilistic upper bound of ice concentration, not a point forecast. **Verified gap in the incumbent:** meshiphi's own IceNet loader ingests forecasts and explicitly discards the uncertainty — its `df.drop(columns=[...])` call removes `sic_stddev` and `ensemble_members`. BAS's own production polar router plans on the mean and throws the spread away. We reinstate it, and our vessel subclass thresholds on the calibrated bound instead of the raw field. Falsifiable: moving the risk slider must produce a different route geometry, not different shading.

D2 — Physics-first drift with a classical-ML residual, and a stated blind spot

Neither an opaque black box nor physics-only. **Honesty commitment displayed in the UI, not just the docs:** BYU/USNIC track only large tabular bergs. Growlers and bergy bits — the ice that actually holes a hull — are neither in the training data nor predicted. Competitors will present iceberg avoidance as solved. It is not.

D3 — Multi-source disagreement surfaced and acted on, not resolved away

Grounded in a documented finding: Copernicus systematically underestimates ice concentration relative to regional charts, especially in the marginal ice zone. A team using one source as ground truth is building on a documented bias without saying so. Where our sources disagree, the system becomes more cautious automatically.

D4 — Bandwidth-honest offline-first design, enforced in CI

The tier boundary is set at “sync environmental state, not answers,” so the master can change waypoints and re-plan with zero connectivity. A CI test **fails the build** if a pack exceeds budget. Demo consequence: disconnect the network mid-demo and continue re-planning routes — unfakeable, and it is how the system is designed to work at sea.

D5 — Corridor- and season-specific, against the real vessel

Built for the actual Cape Town → Bharati/Maitri austral-summer run, not for “Antarctica” in general. Models still train circumpolar because more data is free for a fully convolutional network — only the operational claim is scoped. An MoES/NCPOR audience knows this corridor and this ship; demonstrated fidelity on the real case is verifiable, circumpolar generality is not.

D6 — Honest quantification, including where we decline to quantify

Fuel relative only. Ice thickness disclosed as a literature lookup table. Skill horizon is an output of the evaluation, not an assumption. Bathymetry disclosed as largely predicted. Iceberg drift capped at 72 h. Why this is differentiation and not weakness: competitors will display confident absolute fuel savings and a single authoritative ice number, and every one of those figures is undefendable under questioning. Being the team that already knows which of its numbers are soft is the stronger position — and StormGeo, the actual commercial incumbent, keeps human route analysts in the loop 24/7 precisely because full automation is not trusted at the edges.

D7 — Real prior art reused and extended, with the extensions named

Reuse → adapt → extend → wrap → replace was applied in order, and the reuse candidate survived. We use BAS's own tool and add three specific things it does not do: uncertainty-aware ice thresholding, an iceberg hazard layer through the native `excluded_zones` hook (no fork required — source-checked), and a dominance-filtered candidate route set. A domain-literate judge will know PolarRoute exists. Arriving with “we used BAS's tool and here are three things it does not do that we added” is far stronger than a from-scratch A* that invites the question of why the field's own tool was ignored.

§8 Honesty commitments and known weaknesses

This section exists because the project's central discipline is attacking our own solution before a judge does. Everything here is a deliberate, recorded position.

8.1 Things we refuse to claim

Topic	Our stated position
Absolute fuel figures	Relative between candidate routes only. Absolutes labelled uncalibrated model output, because the fuel polynomial is fitted to a different ship.
Ice thickness and density	A literature-derived climatological lookup table — which is what BAS's own tool uses. Labelled parameterisation, never measurement.
Small ice	Growlers and bergy bits are not tracked by any dataset and are not predicted. Stated in the UI next to the iceberg layer.
Bathymetry	Largely predicted from altimetry, not surveyed. Provenance-conditional depth margin, and a “predicted bathymetry” overlay.
Navigation authority	This is decision support. It does not replace Electronic Navigational Charts, and it says so in the product.
Forecast horizon	The declared skill horizon is an output of the evaluation — the last lead that measurably beats the raw operational forecast — not an assumption.
“Pareto frontier”	We call it a dominance-filtered candidate set from a 12-run sweep, because 12 samples of a continuous surface is a set, not a frontier.
Iridium testing	Resumable delta sync will only ever be tested over throttled localhost. We say “tested under emulated link constraints,” never “tested over Iridium.”

8.2 The riskiest assumption in the whole design

STATED PLAINLY

That the calibrated bound is well enough covered for routing on it to be genuinely safer, rather than merely more conservative-looking. The entire uncertainty-aware differentiation and the safety argument for the routing layer rest on it.

Conformal calibration's coverage guarantee assumes **exchangeability** between calibration residuals and operations — and the post-2016 Antarctic regime is precisely the distribution shift that can break exchangeability. Mitigated by a pre-committed numerical pass/fail rule with a written demotion branch (§5.6), a build-blocking baseline, acceptance on real forecast cycles rather than only the reanalysis proxy, and separate reporting on 2023.

8.3 Climate change is a live threat to this system, and here is the answer

Antarctic sea-ice extent has mostly sat below the 1981–2010 average since **2016**, with record-low summers in 2022 and 2023 and continuing lows through 2025. The winter maximum hit a record low of **16.96 million km² on 10 September 2023**. The published framing is a new sea-ice state which “exhibits different seasonal persistence characteristics, suggesting that the underlying processes controlling Antarctic sea ice coverage may have altered.”

Two consequences that matter:

1. **Persistence — our headline baseline — is itself regime-dependent.** “Different seasonal persistence characteristics” means the +18–31 % is measured against a moving target. Gains must be reported per regime, never pooled across 2015 and 2023 as if they were the same ocean.
2. It sharpens the exchangeability risk in §8.2 directly.

Five defences, in order of how much they actually buy:

1. **The architecture is already the main defence.** We predict a correction to a physics model, not the ice. That model ingests today's assimilated ocean state, so the trend and the regime arrive through the background field, not from the network's memory. Our network only has to learn the model's systematic error pattern — a property of its physics, which drifts far more slowly than the climate.
2. Train on anomalies against a **trailing climatology**, not raw concentration, so a shifting mean does not push inputs out of distribution.
3. **Deliberately hold out the broken years.** Train excluding 2016 and 2023, then test on them. Surviving the two years that broke the field is evidence, not hope.
4. **Drift monitoring in production.** Watch the innovation channel and the residual distribution; when they leave the calibration distribution, widen uncertainty and raise a flag rather than staying confident.
5. **Conformal prediction is the honest failure mode.** When exchangeability breaks — which is what climate change does — coverage degrades visibly and measurably.

THE POSITION TO STATE

“We do not claim immunity to climate change. We claim the system **degrades loudly rather than silently.**” For a safety system that is a stronger and far more credible position than claiming robustness.

§9 Key decisions on record

Twenty-seven Architecture Decision Records are logged in `docs/DECISIONS.md`. Each states the decision, the alternatives, and the reason. Later ADRs openly supersede earlier ones — the reasoning is the deliverable, not just the conclusion.

ADR	Decision
001	Scope to the Cape Town → Bharati/Maitri corridor, austral summer
002	Two-tier architecture, boundary at “sync environmental state, not answers”
003	One codebase, two run modes (<code>MODE=shore vessel</code>)
004	Superseded by 018. Original: 5-member direct multi-lead U-Net ensemble. Retained as the named contingency if the CMEMS data gate fails.
005	The ensemble's uncertainty is a product feature, not diagnostics
006	Superseded by 027. Benchmark against damped anomaly persistence on a strictly temporal split
007	Iceberg drift: Wagner (2017) closed form as the load-bearing baseline
008	Iceberg residual: gradient boosting, and a stretch goal, not baseline
009	Routing: wrap and extend PolarRoute rather than reimplement A*
010	Inject iceberg hazard through PolarRoute's <code>excluded_zones</code> hook; do not fork
011	Superseded by 021. Three-run “Pareto set” — not actually a Pareto set
012	Report fuel as a relative comparison; label absolutes as uncalibrated
013	Use meshphi's literature lookup table for ice thickness/density, and disclose it
014	Surface multi-source disagreement as its own layer instead of picking a winner
015	No LLM, RAG, or agent framework anywhere in this system
016	Demo runs from committed scenario packs, never a live fetch
017	Enforce the sync budget in CI
018	Re-aim the sea-ice model as a bias-correction head on the CMEMS operational forecast (the pivot)
019	Training product and window: NOAA/NSIDC 25 km CDR as the single observation product
020	Calibration by stratified split-conformal prediction with a pre-committed pass/fail rule
021	Multi-objective routing is a 12-run sweep with dominance filtering, named honestly
022	The router always plans the whole voyage; the grade of each leg degrades instead
023	GPU ashore for training, CPU aboard for inference, nothing trained at the nodal centre
024	QC before fusion, one target grid, per-channel imputation, and an obs-only pack type
025	Iceberg drift projection capped at 72 h, with an area-fraction auto-downgrade
026	Depth margin is conditional on bathymetric provenance (GEBCO TID grid)

ADR	Decision
027	Raw packed CEMS is a mandatory, build-blocking evaluation baseline

§10 Current state, open gaps and what happens next

10.1 Two tracks, kept deliberately separate

	SIH prototype — 8 Sept 2026	SIH final — December 2026
Lives in	<code>isih/</code>	repo root — <code>models/</code> , <code>api/</code> , <code>web/</code>
Model	small, single member, short training window	5-member ensemble, 1993-2024
Input channels	12	19
Base field	GLORYS12 reanalysis	real CMEMS forecast archive
Calibration	none — reports raw output	stratified conformal
Deliverable	results, claims and proofs for a slide deck — no UI	full web application, both tiers

The one thing shared is the U-Net architecture file, deliberately not duplicated — so the prototype genuinely demonstrates the production architecture at smaller scale rather than being a different thing wearing the same name.

10.2 What is done

- **DONE** Real data pipeline — 1,096 NSIDC files, GLORYS12 paired, area-average regridded verified
- **DONE** Model trained on Kaggle T4; beats every baseline at every horizon
- **DONE** Routing computed with real PolarRoute on real satellite ice; figure renders with self-computing claims
- **DONE** All architecture documentation — 27 ADRs, differentiation, weakness analysis, competitive analysis, dataset audit
- **DONE** Daily CMEMS harvest cron running (data accumulating for the production training set)
- **DONE** NCPOR overlap check — verified greenfield; NCPOR has no forecasting or route-planning tool

10.3 What is open

Priority	Item
CRITICAL	The route figure's baseline is a strawman. A great-circle straight line is not “existing technology” — no captain sails it. Real practice is a master routing on yesterday's observed ice plus ~100 nautical miles of helicopter reconnaissance. Replace with a ladder: same router, ice input = {observed persistence, raw forecast, our corrected forecast}.
CRITICAL	Route-regret hindcast is unbuilt , and it is the highest-value experiment in the project. Plan on day 0 under each ice input, then replay the voyage through the ice that actually occurred and score real transit time, fuel, and cells the ship could not enter. This is what converts “18-31 % lower error” into days and tonnes. It can come out against us.
CRITICAL	The no-background ablation — ~20 minutes, already written into the notebook. Either kills the leakage objection outright or tells us something we are obliged to disclose.
MAJOR	GEBCO not downloaded — the computed route has no depth constraint at all.
MAJOR	Production code not started: zarr dataset loader, production training loop, ensemble, conformal calibration.
MAJOR	Corridor mesh size and the daily incremental delta are unmeasured — the delta is the load-bearing bandwidth number.

Priority	Item
MINOR	The route figure uses a red/green pair that deuteranopic viewers can struggle to separate. Meaning survives via dash pattern and markers, but a one-line swap removes the risk.
MINOR	MV Vasily Golovnin's published dimensions and ice class not yet sourced — needed for the vessel subclass and the depth margin.

10.4 New findings from the navigation review (1 September 2026)

A design review against the full problem-statement text surfaced three material gaps, all verified from primary sources or installed source code. Full detail in `docs/NAVIGATION_RESEARCH.md`.

(a) The vessel ice limit should be IMO POLARIS, not a number we chose

The prototype uses `max_ice_conc: 80` — invented. The IMO already mandates a framework: **MSC.1/Circ.1519, the Polar Operational Limit Assessment Risk Indexing System (POLARIS)**, referenced by the Polar Code.

$$RIO = (C_1 \times RIV_1) + (C_2 \times RIV_2) + \dots + (C_n \times RIV_n)$$

C_i concentration of ice type i , in tenths

RIV_i Risk Index Value for that ice type at THIS SHIP'S ice class

$RIO \geq 0$ normal operation

$-10 \leq RIO < 0$ elevated risk — speed capped (PC1 11 kt · PC2 8 kt · PC3–PC5 5 kt · below PC5 3 kt)

$RIO < -10$ avoid in voyage planning

- Turns an invented threshold into a number a judge can look up.
- Makes the ship's ice class a real input that changes the answer — exactly what the problem statement asks for.
- Makes risk **continuous rather than binary**, so the router can prefer safer ice instead of only avoiding a cliff edge.
- **§1.7.3 requires a safe stand-off distance from glacial ice** — the iceberg safety buffer, as a regulatory requirement rather than an invented one.
- **§1.6.4 allows adding +10 to the RIO along an icebreaker's broken track** — the formal mechanism by which a following vessel benefits from a leader, and the basis for any convoy or fleet-ordering model.

AND IT FORCES A MODEL UPGRADE — IN A GOOD WAY

RIO is indexed by ice **type**, not concentration, and WMO stage-of-development maps to thickness (new <10 cm · grey 10–15 · grey-white 15–30 · thin first-year 30–70 · medium 70–120 · thick >120 cm). So **POLARIS cannot be computed from a concentration forecast alone — it requires a thickness forecast too**. GLORYS12 carries `sithick`, and the U-Net extends to a second output channel with no architectural change. This is not scope creep; it is a model requirement falling out of the regulation.

(b) PolarRoute's router has no time dimension — verified by source inspection

Searching the installed package for time-dependent mesh handling returns nothing in the route planner; `datetime` appears only as a running sum over output legs. Mesh time bounds select and aggregate a slice; they do not produce per-timestep layers.

The router therefore assumes the environment is frozen for the entire voyage — 17.4 days, in our own computed route. The problem statement explicitly requires dynamic recalculation as ice layers

move. This is the gap that makes a forecast usable: a static router structurally cannot consume a lead-time forecast, because it has no notion of “the ice you will meet when you actually arrive.” The extension is a **time-expanded graph** — a mesh stack, one layer per forecast day, with edges (cell, t) → (neighbour, t + traveltime).

(c) Wave-added resistance is dead code

`wave_resistance()` is defined in three `PolarRoute` vessel classes and called from none of them. Resistance sums only wind and ice. Waves act solely as a binary accessibility cutoff, so there is **no sea-state fuel penalty anywhere** — including the Roaring Forties, where most of the fuel is actually burned. A latent bug compounds it: the dead method reads `params["Beam"]` while the constructor sets `"beam"`. Note that a planned vessel subclass “with wave resistance” will silently do nothing unless `model_resistance` is also overridden.

(d) One logistics fact that must not go on a slide wrong

Indian expeditions are flagged off from Goa, but personnel fly to Cape Town and the sea leg starts there. Do not present “Goa → Antarctica by sea” as current practice — an MoES judge will know. The credible forward-looking framing is that **GRSE signed an MoU with Kongsberg in June 2025 to build India's first indigenous Polar Research Vessel** at Kolkata for NCPOR; when India has its own hull, the voyage genuinely starts from an Indian port.

10.5 The phase plan to December

Phase	Duration	Contents and exit condition
0 — spine	days	Repo, pinned environment, FastAPI skeleton, React shell. Plus two things that cannot be deferred: the daily CMEMS harvest cron, and the NCPOR overlap check. Both complete.
1 — real routing on real data, and the data gate	~1 week	GEBCO + TID, NSIDC, GFS ingestion with QC; corridor mesh; vessel subclass; single-objective routing. No ML yet — deliberately gets a genuinely useful real-data product working before any model exists. Also owns four measurements and the CMEMS go/no-go gate.
2 — the trained model	1–2 weeks the long pole	Historical ingestion; residual U-Net training (8–12 GPU-hours); 5-member ensemble; conformal calibration; the full evaluation protocol including the build-blocking baseline and the coverage gate.
3 — uncertainty and hazards	~1 week	Calibrated bound layer; overridden ice threshold; Wagner drift with the 72 h cap; berg exclusion zones; the 12-run sweep; fusion disagreement layer; horizon tiering and climatology mode. Exit: the risk slider visibly moves the route, and its label matches measured coverage.
4 — offline tier	days	Pack build/sign/delta; vessel mode; staleness and confidence UI. Exit: full re-plan with networking disabled.

STATED FALLBACK IF PHASE 2 SLIPS

Ship Phases 1 + 3 + 4 with **raw packed CMEMS as the routed field** and inter-source disagreement as the uncertainty proxy. The pack, the mesh, the slider mechanics, the berg layer and the offline demo are all unchanged; the pitch degrades from “our model improves the operational forecast” to “our decision architecture makes the operational forecast routable.” The one thing that must survive is a trained model — so if the deep correction slips, the **iceberg residual GBM is promoted from stretch to baseline**, because it trains in minutes on data already downloaded. This ordering is decided now so it is a plan, not a scramble.

10.6 What is prepared before travel versus built at the nodal centre

Frozen on USB before travel	Built or run at the 36-hour finale
Trained weights + conformal calibration tables	Live re-planning demos, scenario walkthroughs
Evaluation results and all figures	UI polish, presentation assembly
Committed demo scenario packs	Judge-requested re-runs on committed packs (real computation, ~9 s + ~84 s sweep)
Environment pinned and pre-installed	Bug fixes, integration of anything left open

Nothing on the left is regenerated on site. Nothing on the right needs a GPU or a network. “Re-run training at the nodal centre” is not a thing.

§11 Repository map and how to get started

11.1 Where everything lives

SIH26059-antarctic-navigation/	
├ isih/	ISIH PROTOTYPE (8 Sept) – kept separate on purpose
│ └ RESULTS.md	← the numbers to quote, with caveats
│ └ download_nsidc.py	fetch observed sea-ice concentration (no auth needed)
│ └ grid.py	NSIDC grid definition + GLORYS12→NSIDC area-average regridding
│ └ build_dataset.py	pair background with truth; nothing imputed, missing days dropped
│ └ features.py	the 12-channel builder + the baselines the model must beat
│ └ make_route_figure.py	compute the real route with PolarRoute
│ └ plot_route.py	render the figure; computes its own claim numbers at render time
│ └ kaggle/	self-contained training notebook (sources embedded)
│ └ figures/	route_map.png, mesh.json, routes.json
├ models/	
│ └ sic_correction/unet.py	THE SHARED ARCHITECTURE – used by prototype and production
├ docs/	the project's working memory
│ └ ARCHITECTURE.md	system design, rev 2
│ └ ML_ARCHITECTURE.md	every model, rev 2 – the densest document
│ └ DATASET.md	every data source, verified live
│ └ DECISIONS.md	27 ADRs
│ └ DIFFERENTIATION.md	D1–D7, no adjectives allowed
│ └ WEAKNESS_ANALYSIS.md	the adversarial review rev 2 answers
│ └ COMPETITIVE_ANALYSIS.md	what already exists, and where competitors will be shallow
│ └ GAP_ANALYSIS.md	method survey
│ └ NAVIGATION_RESEARCH.md	POLARIS tables, PolarRoute gaps, Indian logistics (new)
│ └ backlog.md	every open item as a one-liner – READ THIS FIRST
│ └ PRIMER.md	domain background
│ └ VIVA_PREP.md	expected questions and answers
│ └ TEAM_GUIDE.md	working agreements
│ └ DEMO_PLAN.md	how the demo runs
└ .github/workflows/	
│ └ cmems-harvest.yml	daily 06:00 UTC – DO NOT SWITCH OFF

11.2 If you are new, read in this order

1. This document, §1 and §2.
2. docs/backlog.md — under two minutes, and tells you everything still open.
3. isih/RESULTS.md — the real numbers and their caveats.
4. docs/ML_ARCHITECTURE.md §1 — the model, in full depth.
5. docs/DIFFERENTIATION.md — why this is not the obvious build.

11.3 Reproducing the results

- **Training:** open isih/kaggle/isih_train.ipynb on Kaggle with Internet On, Accelerator GPU T4 ×2, and CMEMS_USERNAME / CMEMS_PASSWORD in Kaggle Secrets. It downloads its own data. Roughly 1–2 hours.
- **Routing:** needs Python ≤3.11 — PolarRoute will not build on 3.12+. Create the environment with `uv venv --python 3.11`, then run isih/make_route_figure.py followed by isih/plot_route.py.
- **After editing** grid.py, features.py or unet.py, regenerate the Kaggle notebook — it embeds those sources.

11.4 Working rules for the team

- **No dummy data, ever, in the product.** Unit-test fixtures are the only exception.
- **Deferred work never lives only in chat.** If you find something and do not fix it, it goes into `docs/backlog.md` as a one-liner with a pointer to the evidence.
- **Never trust code because it looks right.** Open the browser, run the query, evaluate the model on real held-out data.
- **Attack our own solution before a judge does.** Every number in this project that looked good was re-derived until it either survived or was withdrawn.

§12 Glossary

ADR — Architecture Decision Record. A short note stating a decision, its alternatives, and why.

AMSR2 — Advanced Microwave Scanning Radiometer 2, a Japanese satellite sensor giving ice concentration at 12.5 km.

Beset — a ship stuck fast in ice, unable to move under its own power. The failure mode this system exists to prevent.

Bergy bit / growler — small pieces of floating ice, roughly car- to house-sized. Not tracked by any dataset; the ice that actually holes a hull.

CDR — Climate Data Record. A long, consistently processed satellite record suitable for training.

CMEMS — Copernicus Marine Environment Monitoring Service. The EU's operational ocean forecasting service; our base field.

Conformal prediction — a distribution-free method that turns model outputs into prediction intervals with a measured coverage guarantee.

CRPS — Continuous Ranked Probability Score. Scores a probabilistic forecast, not just its mean.

Dijkstra — a shortest-path algorithm that is exact on a graph, as opposed to a heuristic search.

Exchangeability — the statistical assumption that calibration data and operational data come from the same distribution. Climate change breaks it.

GEBCO — General Bathymetric Chart of the Oceans. The open global seabed-depth grid.

GLORYS12 — CMEMS's global ocean reanalysis at 1/12°, 1993–present. Our training background field.

IIEE — Integrated Ice Edge Error. Measures how wrong the position of the ice edge is, in km² — the navigationally meaningful metric.

Innovation — in data assimilation, the difference between a model's forecast and the observation. Our key input channel.

Iridium — a low-Earth-orbit satellite constellation, the only communications that genuinely covers the poles. Certus peaks around 704 kbps.

ISEA — Indian Scientific Expedition to Antarctica, run annually by NCPOR.

Lead (ice) — a navigable crack or channel of open water through pack ice.

Lead (time) — how many days ahead a forecast is valid for. Both senses appear in this project; context distinguishes them.

MIZ — Marginal Ice Zone. The transitional band between open water and solid pack ice, roughly 15–80 % concentration. Where ships sail and where forecasts are worst.

MoES — Ministry of Earth Sciences, Government of India.

NCPOR — National Centre for Polar and Ocean Research, Vasco da Gama, Goa. The problem statement's owner.

NSIDC — National Snow and Ice Data Center, USA. Source of our truth field.

Persistence — the baseline forecast “tomorrow will look like today.” Deceptively strong at short lead, and the baseline that matters.

Polar Class (PC1-PC7) — IACS structural ice classes for ships, PC1 strongest. Finnish-Swedish classes IA Super / IA / IB / IC are the lighter equivalents.

POLARIS — the IMO's Polar Operational Limit Assessment Risk Indexing System. Converts ice conditions plus ship ice class into a go / caution / avoid outcome.

Polynya — a persistent area of open water surrounded by sea ice.

Reanalysis — a retrospective reconstruction of past conditions using all observations, including ones from after the moment being reconstructed. Not a forecast.

Residual / bias correction — learning the error of an existing model rather than the quantity itself.

RIO / RIV — Risk Index Outcome and Risk Index Value, the two numbers POLARIS is built from.

RMSE — Root Mean Square Error. Lower is better.

SAR — Synthetic Aperture Radar. Sees through cloud and polar darkness.

SIC — Sea Ice Concentration, the fraction of a grid cell covered by ice, 0 to 1 (or 0–100 %). The central quantity in this project.

SIT — Sea Ice Thickness. Much harder to measure than concentration, and required for POLARIS.

U-Net — a convolutional neural network shaped like a U: it compresses an image, then expands it back, with skip connections preserving fine detail. Standard for image-to-image prediction.

Voyage pack — our compressed, signed, versioned bundle of everything the ship needs. Under 1 MB.

Wagner (2017) — the closed-form analytical iceberg drift model that is our physics baseline.

FINAL NOTE

If any statement in this document cannot be traced to a measurement, a primary source, or a line of code, treat that as a defect and raise it. The project's entire credibility strategy rests on the difference between what we have measured and what we merely believe — and on saying which is which.